A Cross-Attention-Based Multimodal Framework for Parkinson's Disease Detection

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report submitted in partial fulfillment of the requirement for the award of degree of

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Branch: COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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SYNOPSIS

Parkinson's disease is a disorder of the nervous system which causes difficulty in moving, speaking, hand writing and walking. Early detection can aid in timely clinical evaluation. But, the traditional assessments are based on the clinical exam, and a single-modality approach can miss critical symptoms. Therefore, a multimodal approach is needed to analyze complementary behavioral patterns.

This project aims to develop a multimodal, uncertainty-aware deep learning approach for Parkinson's disease screening using speech, handwriting and gait modality. It is based on the MDVR-KCL dataset of speech, the PhysioNet Gait dataset of gait and the NewHandPD dataset of handwriting. The modality-specific features are extracted using a Wav2Vec2, a Vision Transformer (ViT) and a 1D Convolutional Neural Network (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) network, which are then used for the classification of the samples as Parkinson’s disease or healthy control.

To increase the reliability of the predictions, methods like predictive entropy, mutual information, temperature scaling, and split conformal prediction are investigated to measure the confidence and uncertainty. An evidential and cross-modal reliability-based fusion mechanism fuses the predictions with consideration to their reliability across the modalities. To overcome the absence of common individuals in all three modalities, the fusion strategy is developed.To resolve the problem of the lack of common individuals in all three modalities, the fusion strategy is developed. Performance is assessed based on accuracy, precision, recall, F1 score, ROC-AUC and the appropriate uncertainty and probability-quality metrics.

The proposed system can also achieve the goal of ensuring the reliability, interpretability, and multimodality of screening for Parkinson's disease with complementary behavioral information and prediction uncertainty.

CHAPTER 1
INTRODUCTION

1.1 Background and Motivation

Parkinsons disease is a brain disorder that changes how a person moves. It can cause movements, changes in speech and trouble walking. Each person experiences levels and mixes of symptoms making Parkinsons disease hard to spot. Finding Parkinsons disease early lets doctors assess quickly and give the treatment. Traditionally doctors find Parkinsons disease by looking at the patient asking about history and doing a brain exam. These steps are important. They need a specialist and do not always give an objective, automated look at small behavioural changes.

1.2 Need of different modalities

Three digital biomarkers each show a different part of Parkinsons disease. Speech signals may show changes in voice volume how long words are held, rhythm and clarity. Handwriting patterns may show movements, uneven strokes and changes in how writing is done. Gait signals may show changes in walking patterns, timing and coordination. Looking at each biomarker gives useful information but each only shows part of the behavioural changes in Parkinsons disease. Therefore combining data from speech, handwriting and gait can build screening systems.

1.3 The Challenges with Multimodal Systems

Using multimodal systems has challenges. Models can give predictions with levels of confidence. Input quality also varies – speech recordings may have background noise handwriting may look unclear. Gait signals may be incomplete. If all predictions are treated as reliable an uncertain prediction can hurt the final decision. Public datasets for speech, handwriting and gait are collected separately. It is hard to get all three from the same person. This project proposes an uncertainty-aware Parkinsons disease screening system that builds separate deep-learning models for speech, handwriting and gait checks how reliable each prediction is and looks at ways to combine the outputs.

1.4 System Description

The proposed system has these parts: modules that process each modality, a module that checks uncertainty and reliability fusion part and an explainability part.

Speech Analysis Module:

The speech module uses the MDVR-KCL Speech Corpus to train a model on speech. The model uses Wav2Vec2 to turn speech into a representation then decides the class. Calibration and uncertainty analysis estimate how reliable the model's predictions are.

Handwriting Analysis Module:

The handwriting module takes data from the NewHandPD dataset. Uses the ViT model to learn visual features and classify images as Parkinsons disease or healthy.

Gait Analysis Module:

The gait module uses the PhysioNet Gait in Parkinsons Disease dataset, which has walking signals. It uses a CNN–BiLSTM architecture to find local patterns and dependencies for classifying gait.

Uncertainty and Reliability Analysis Module:

This part looks at entropy and mutual information to measure epistemic uncertainty. It applies temperature scaling to check probability calibration. It explores split conformal prediction to build prediction sets with a target coverage.

Cross-Modal Fusion Module:

The fusion module combines the predictions or evidence from the speech, handwriting and gait models while considering their estimated reliability. Because the datasets do not usually have samples from the patients the system uses an unpaired setting and evaluates within it.

Explainability Module:

The explainability part uses methods like SHAP and when possible Grad-CAM to explore the input features or regions that contribute to the model predictions.

1.5 Scope of the System

The project scope covers training deep-learning models for speech, handwriting and gait studying reliability and uncertainty for all three checking probability calibration and conformal prediction building a fusion algorithm that works with the available data applying explainable AI techniques and measuring performance, calibration and uncertainty, with proper metrics. This project does not aim to create a diagnosis or replace a neurological examination. The performance of the approach created by this project depends a deal on how well the individual models work, on the characteristics of the dataset used and on how valid the fusion algorithm applied is.

1.6 Problem Statement

There are many problems, with current machine learning solutions that try to detect disease. These solutions usually concentrate on one type of input or they try to mix several types of input to get better overall performance yet they do not consider how reliable the predictions are. As a result even if different behavioural patterns might help each other the predictions made by the models can end up being inconsistent calibrated or even unreliable. The problem statement of this project is to build a Parkinsons disease detection system that's aware of uncertainty and is explainable. This system will analyse speech, handwriting and gait separately. It will also measure how reliable each prediction is and it will investigate the way to fuse the different modalities together.

CHAPTER 2

LITERATURE STUDY

2.1 Overview

The most commonly used methods of analysing speech, handwriting and gait are used to study Parkinson's disease. All of these are symptoms of Parkinson's: Change in voice, Fine-motor skills, Walking pattern. Preliminary studies have demonstrated it is feasible to apply machine learning and deep learning to glean useful patterns to enable automatic Parkinson's detection from these data. The method, however, is highly feature dependent, and highly dependent on the data used, the model structure and the method used to predict the confidence.

2.2 Speech-Based Approaches

The illustrative work done by Narendra et al., (2021) and Rey-Paredes et al., (2025) for the speech based technique for detection of PD is very illustrative. In this paper, Narendra et al. delve into the effect of Voice-Source and Acoustic features on Parkinson's disease (PD) diagnosis using Support vector machine and End-to-End techniques. The study highlights clearly the importance of the different voice parameters in the recognition of the disease; however, in this study, this is applicable to only one motor related modality – speech. Later, Rey-Paredes et al. (2025) added two data augmentation methods based on deep learning and generative adversarial network (GAN) to the Parkinson's disease detection from raw voice wave form. This not only gives a guideline to decrease engineered features but also to overcome the lack of data by data augmentation, but only considers the speech modality.

In recent years with the introduction of the pretrained speech representation, another opportunity has emerged to improve the speech analysis with automatic detection of PD. For example, Klempir et al. (2025) used the Wav2Vec 1.0 and Wav2Vec 2.0 embeddings to detect PD from a variety of speech modes and languages. Their project has certainly demonstrated that speech representation can be beneficial to

retrieve information about disease from voice recordings. Also, Speech models approach cannot detect the motor abnormalities present in a PD patient which is one of the characteristic features of the PD patient, thus studying it using Speech models as one of the modalities for the detection of the disease.

2.3 Handwriting-Based Approaches

Another method of determining the presence of Parkinson's disease is by handwriting analysis. Allebawi et al., (2024) investigated the online handwriting in relation to the beta-stroke segmentation, beta-elliptical features and fuzzy perceptual detector. On the other hand, the handwriting approach can prove to be a good way to identify motor-related abnormalities, where handwriting designed features are very much used, and handwriting data is scarce. Therefore, it could be feasible to adopt deep-learning image representations which are able to learn the relevant patterns directly from handwritten images.

2.4 Multimodal Approaches

Combining information from different sources is possible using multimodal assessments. Vásquez-Correa et al. (2019) came up with the idea of a deep learning model including the features from the patient's speech, hand writing and gait to perform patient assessment for PD. In this research the use of more than one modality (not one single type of data) is accentuated. However, there is much additional work to be done with regard to modality specific models, explicit uncertainty estimation and decision making with uncertainty. In addition, agreement and uncertainty on the prediction reliability should be carefully evaluated for using multiple models to generate multimodal predictions.

2.5 The Dataset Availability Problem

The other critical one is there's a set of data with a lot of modalities and a particular patient. Since the mode of recording (speech, handwriting and gait) is recorded one at a time, learning and per patient evaluation is not possible. So, when it comes to data multimodal fusion, the data set's structure and feature should be considered.

Important: If all modalities are used for one patient, it is not possible to evaluate all the modalities.

Literature to be summarized here will be literature that has been reviewed in this literature review. Studies in this area have shown that there is complementary information in the speech, handwriting and gait that may be beneficial for the diagnosis of PD. Acoustic features, pretrained speech representations, handwriting analysis and multi modal deep learning are the aforementioned methods that have been proven to be helpful. In this case, a system should be designed that incorporates the modality-specific deep learning models, uncertainty estimation, probability calibration, conformal prediction and cross-modal reliability fusion.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction to System Analysis

System analysis allows one to define all required functional capabilities and quality requirements of the proposed Parkinson's disease detection system. Specifically, the system will be able to process data related to speech, handwriting, and gait modalities with the use of modality-specific models and combine them via cross-modal reliability fusion. Uncertainty, calibration, and consistency are the factors that will make predictions more conservative and interpretable.

3.2 Functional Analysis

3.2.1 Input Processing

The system has to support the processing of three input modalities: recordings of people's speech, images of handwriting, and time-series gait data. The appropriate preprocessing pipeline has to be developed to transform raw input into a suitable format for a particular modality-specific deep-learning model. The speech data are processed by the Wav2Vec2 model, handwriting images by the Vision Transformer , and gait time-series data by 1D CNN with BiLSTM.

3.2.2 Modality-Specific Predictions

The output of each modality-specific model is the prediction or the representation that can be used in the fusion module. In addition, the system has to provide an estimation of the probabilities of Parkinson's disease and of the healthy class in accordance with the provided modality output. The system has to be able to compare predictions of different modalities and find the areas where there is agreement, disagreement, or an uncertain prediction.

3.2.3 Uncertainty Estimation

The uncertainty estimation module estimates the uncertainty inherent in predictions of the models. Various methods such as predictive entropy, evidential uncertainty estimation, and deep ensembles can be implemented depending on the pipeline chosen. Temperature scaling allows adjusting probabilities estimated by the model according to the validation data, while conformal prediction produces a set of possible classes for the input.

3.2.4 Cross-Modal Reliability Fusion

The cross-modal reliability fusion module combines the information received from each individual modality. This module evaluates the consistency of predictions, takes into account information about uncertainty and conformal prediction, and applies specific decision rules to see if the prediction can be accepted or not. Such an approach is required in order to make sure that the system does not rely on a single modality in case of contradictory or insufficient evidence.

3.2.5 Explainability

Explainability should also be incorporated in order to enhance the interpretability of the predictions. For example, SHAP can be used to explain the contribution of various features, and for image-based predictions Grad-CAM or appropriate attention visualisations may be considered. However, these techniques only be included in the final system after they have been implemented and their outputs have been validated.

3.3 Non-Functional Analysis

3.3.1 Reliability and Consistency

The system needs to reliably and consistently process speech, handwriting, and gait data. For each modality, suitable preprocessing and model evaluation should be carried out in order to reduce errors resulting from differences in input formats, signal quality, or data distributions. It is important that the system has a modular architecture so that individual models can be trained, tested, and updated separately without

requiring substantial modifications to the whole pipeline.

3.3.2 Performance

It is important to consider performance since the system includes a number of deep-learning models as well as extra uncertainty and fusion components. In order for the framework to be practical for software-based experimentation on the available computing resources, efficient preprocessing, model inference, and effective use of resources are required. The adoption of pretrained representations and modular pipelines can aid in managing the complexity of the implementation.

3.3.3 Evaluation Metrics

To evaluate reliability, appropriate classification metrics should be used, such as accuracy, precision, recall, the F1-score, and ROC-AUC where relevant. Probability calibration should be assessed by means of metrics including Negative Log-Likelihood (NLL), the Brier Score, and Expected Calibration Error (ECE). Conformal prediction should be evaluated in terms of empirical coverage and prediction-set size. These measures offer complementary information regarding classification performance, probability reliability, and uncertainty.

3.3.4 Maintainability and Reproducibility

Maintainability is achieved by separating the models specific to each modality, the uncertainty estimation, and the fusion logic into different components. This approach allows the various modules to be altered or replaced without the need to redesign the whole system. Moreover, reproducibility is important, and this requires that the preprocessing be consistent, that the dataset splits be documented, that experimental setups be fixed where appropriate, and that the results of the evaluation be clearly reported.

3.3.5 Data Availability Constraints

In multimodal research on Parkinson's disease, data availability represents a major restriction. When the speech, handwriting, and gait datasets include different

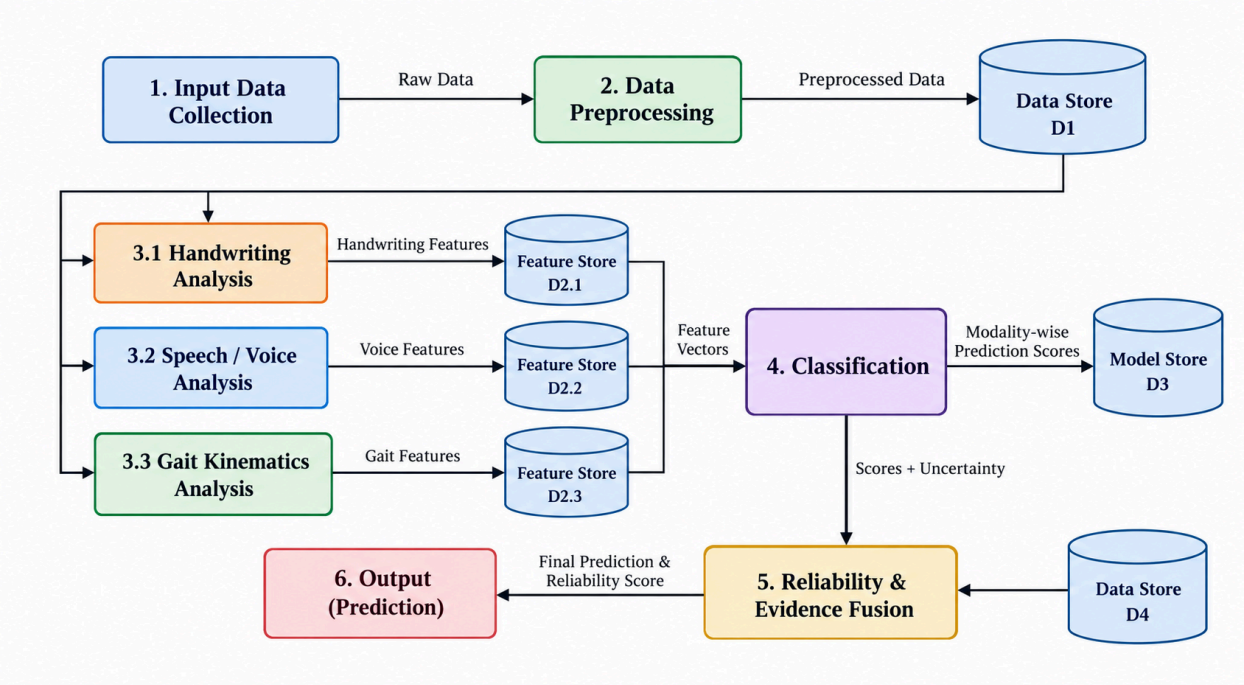
individuals, the system should not assume that the corresponding samples come from the same patient. The fusion method must therefore be assessed in light of the actual data-pairing approach, and the results from simulated or class-consistent unpaired fusion should be given separately from those obtained from a patient-level multimodal analysis.

3.3.6 Research Tool, Not Clinical Diagnosis

In the end, the system should be designed to serve as a tool for detection and to support research rather than as a substitute for clinical diagnosis. Its results need to convey uncertainty and should highlight cases that call for further evaluation. The system must not present its predictions as definite medical conclusions, especially in situations where its confidence level is low, the different modalities give conflicting results, or the input data differ from that which was used during training.

CHAPTER 4

SYSTEM DESIGN



The Data Flow Diagram shows how the data is supposed to flow throughout the proposed Parkinson's Disease Detection System from data collection to prediction of the disease. It represents the relationships among the inputs, the pre-processing, modality specific analysis, classification, data storage and reliability assessment.

4.1. Input data collection and preprocessing

The system captures handwriting images, the voice and gait information from the user and/or from appropriate data source. The data thus collected is then verified and preprocessed in order that it will be in an appropriate format for further analysis. It can contain some type of noise removal, missing value filling, normalisation, image or signal transformation, depending on the type of input data.

4.2 Modality-Specific Analysis

Hand writing, voice analysis and gait analysis modules are used in the system for analysis. Handwriting analysis is based on visual characteristics, voice analysis is based on acoustic characteristics, and gait analysis is based on movement-related characteristics. Each of the modules transforms its input into a suitable feature representation which can be used by the classification model.

4.3 Classification and Reliability Assessment

Classification module enables to apply a machine learning model or deep learning one, trained on the extracted features, to produce the prediction scores. The reliability module weighs in the predictions that are available, the evidence fusion module then evaluates these and fuses them together using the chosen fusion strategy, thus creating a single prediction with the uncertainty taken into account as well as the differences within the predictions made by each modality.

4.4 Data Storage

The preprocessed inputs are stored in one data-store, features extracted from preprocessed inputs are stored in one data-store, trained models are stored in one data-store and prediction results are stored in one data-store. It can then access the information at the appropriate time as well as keeping a structure for the various steps of the processing. Moreover, the results and logs stored can be utilized for model evaluation, debugging and future enhancements.

4.5 Output Generation

The module which produces output gives the final prediction to the user in a comprehensible way. This output can come from a predicted class, a probability (or confidence score) and details about the model's reliability, depending on the model being implemented. The results will give some initial analytical background information but should not be used in place of a clinical examination or diagnosis made by a competent health care provider.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System overview

This project is essentially a clinical decision support system for detection of Parkinson's Disease not only by predicting it but also by stating the confidence on its prediction. Most PD classifiers you'll find in the literature returns only one confidence number coming from a softmax layer and it is well known that the number is not so accurate when the input data are corrupted or when they are coming from a hospital that the model is never exposed to before. We designed our system to cure that issue.

Three fully independent models were created: one for Gait, one for Voice, one for Handwriting. Each model is not only a classifier, but an evidential classifier, with a measure of the evidence it has gathered in favor of each class; this evidence can be used to inform a principled uncertainty score. These three separate decisions are then fed to a central fusion engine, which merges them together into one clinical decision: ACCEPT or FLAG.

Everything is in one inference pipeline, and one Streamlit interactive dashboard, where a clinician can upload files from any of the three modalities and receive a full uncertainty-aware report in real-time.

5.2 Data Preprocessing

In each modality, it was necessary to preprocess the data for use in the analyses, since raw data were very different across modalities; however, the principle for each modality was the same – there was no patient level data leakage whatsoever. That's something I'm very vigilant about when implementing, it's a very easy thing to do in published PD papers to just put a patient on both the train and the test—without realizing it. That is NOT allowed.

5.2.1 Gait (VGRF) preprocessing:

The PhysioNet gait files are tab-separated text files with cryptic file names. We use regular expressions to extract the study code, the group (control vs. patient), subject ID, and walk number from those filenames. Of the 19 original columns, only the 16 VGRF channel readings, left foot (L1-L8) and right foot (R1-R8) are retained, the two total-force columns and the time column are discarded as the total is simply the sum of all channels and would introduce redundancy. Any NaN or infinity values due to faulty sensor measurements are set to zero. Normalization is applied to the training recordings channel-wise, and the statistics are frozen, so that validation and test data are normalized using the frozen statistics and not their own.

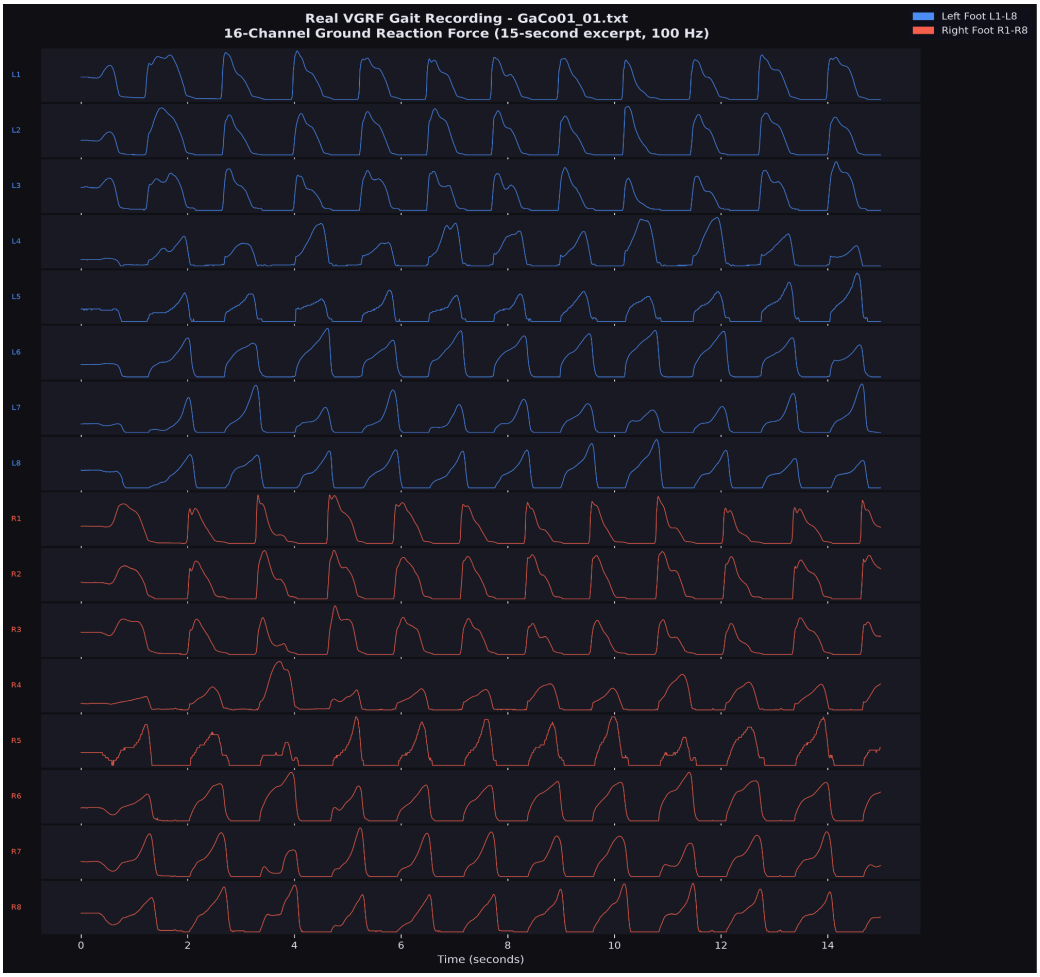


Figure 5.1: Raw VGRF Gait Signal

Short recordings are then zero-padded to window size and continuous recordings are chopped into 5s windows of 500 timesteps with 250 timesteps of overlap (a 500 timestep stride). The last tensor's shape is (500, 16). There is a split itself stratified at the subject level (70/15/15), and there are hard claims that no subject is seen in more than one split.

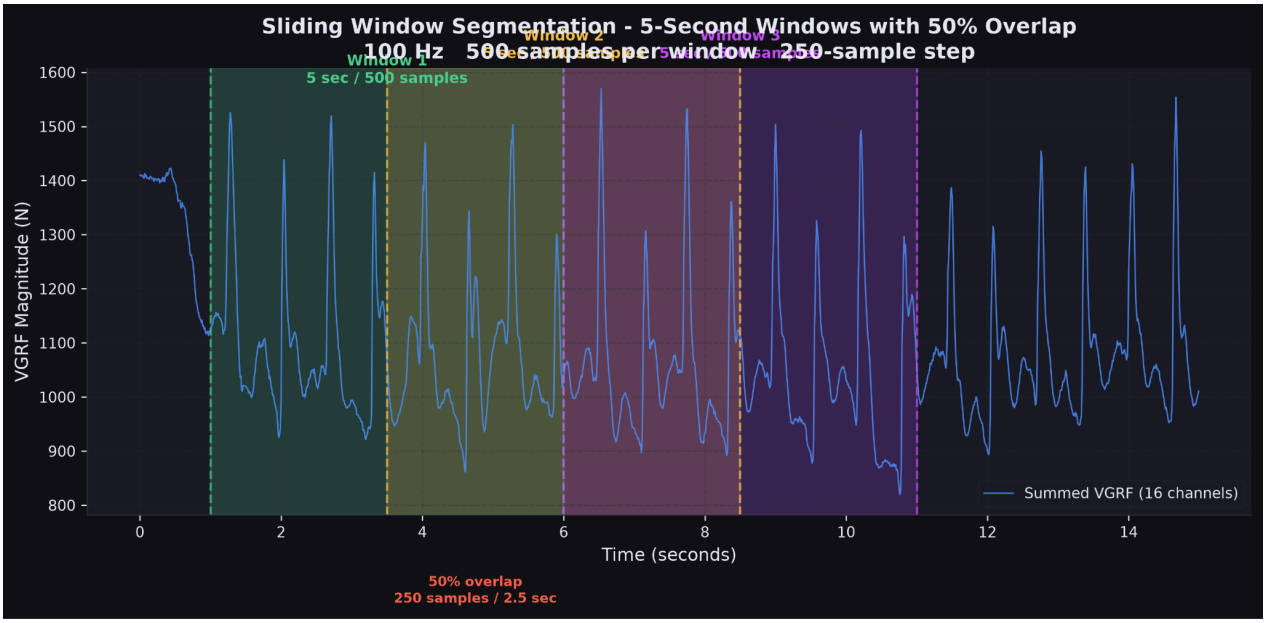


Figure 5.2: Sliding Window Segmentation

5.2.2 Voice preprocessing:

The MDVR-KCL audio files are discovered by scanning the ReadText and SpontaneousDialogue folders for HC and PD subdirectories, and subject IDs are pulled out of the filenames with a regex. Multi-channel audio is averaged down to mono, integer formats are normalized to float32 in the range [-1, 1] and all is polyphase-resampled to 16 kHz with gcd-based up/down factors, resulting in a much higher fidelity than naive resampling. The audio is then divided into 10-second windows (160,000 samples) and zero-padded to the size of one of these windows for short files, while also having 2-second overlaps between windows. Eventually each chunk makes it through the mean zero + unit variance normalization layer that the pretrained model expects to see. Again, stratified splits at the subjects level.

5.2.3 Handwriting preprocessing:

The NewHandPD dataset is packaged as two files: a CSV file for spirals and a CSV file for meanders which point to the location of the images in four folders. We load both manifests and concatenate them together, and resolve each image name to a real file path and mark (and exclude) all the missing files. This split is done using StratifiedGroupKFold, with patient IDs as the grouping variable, which is a better way to ensure patient-level separation across 10 folds than using a single stratified split. Images are cropped randomly to 85–100% of the original size, rotated by ±10° and flipped horizontally, and then normalized to ImageNet means and standard deviations, during training. At test time, augmentation is disabled and the image is only resized to 224×224.

5.3 Model Architectures

5.3.1 Gait — Gait CNN BiLSTM:

This model was built from scratch as gait VGRF is a continuous multivariate time series and there is no off-the-shelf pretrained backbone that fits. The architecture begins with two 1D convolutional blocks with 16 channels first, then 64 channels, and 5 kernel size, which is 50ms at 100Hz (approx. one heel-strike window), followed by batch normalization and ReLU and finally max pooling (64 channels). The convolutional features are then sent to a 2-layer bidirectional LSTM with 128 hidden units in each direction, which models stride periodicity and long-range temporal dependencies in both directions. Global Temporal Mean Pooling then yields a 128 dimensional embedding that is fed into a shared Evidential Head: a dropout layer, a linear layer and a Softplus activation to ensure that the evidence values are non-negative. The output is a 2D vector e_HC, e_PD.

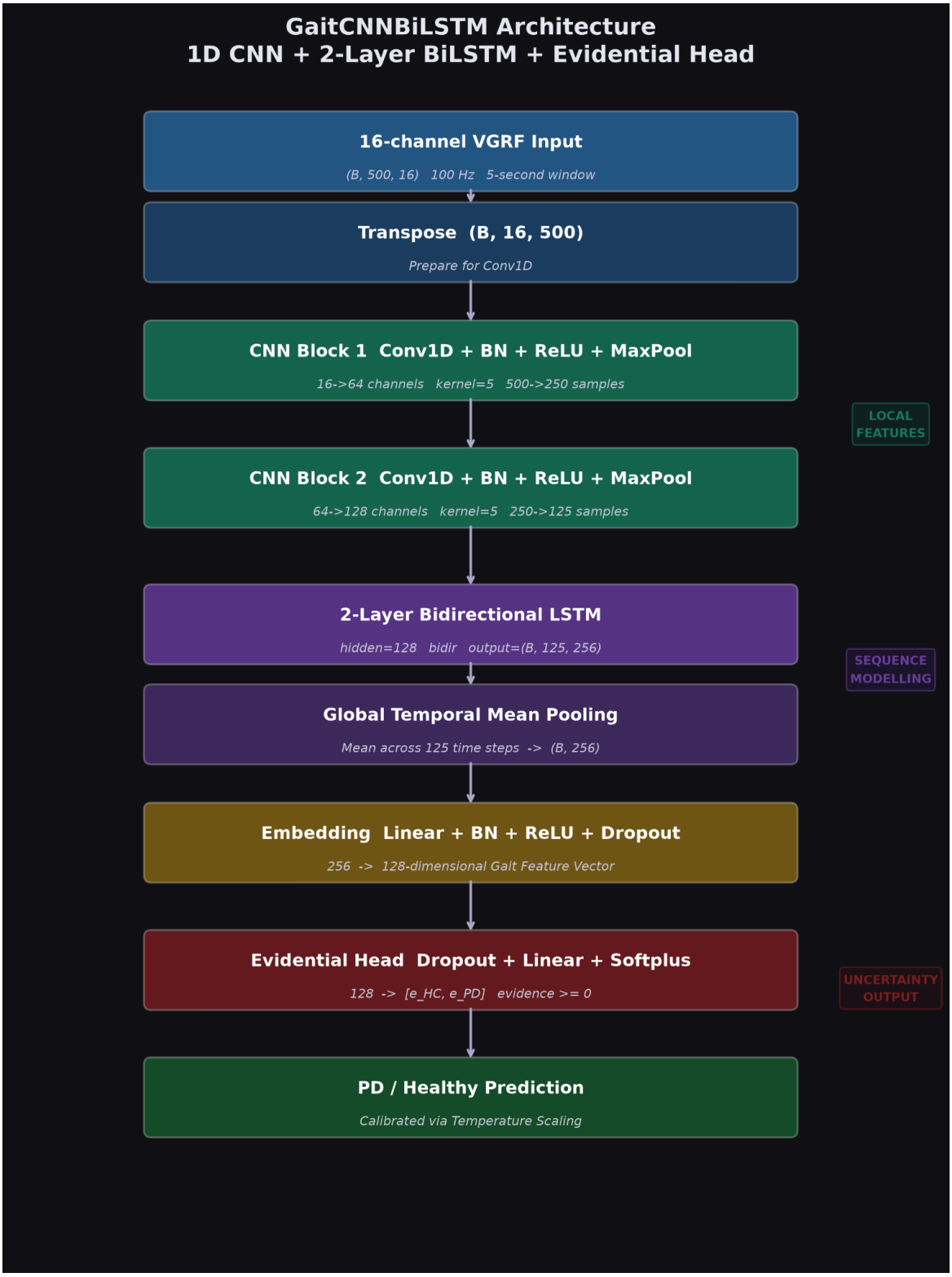


Figure 5.3: GaitCNNBiLSTM Architecture

5.3.2 Voice — Wav2Vec2 For Parkinsons — 5.3.2.

Here we use the self-supervised Wav2Vec2 model (facebook/wav2vec2-base, 94M parameters), which was pretrained on massive amounts of unlabeled speech data and has acquired good acoustic representations. We use its transformer output and then apply “masked temporal mean pooling” along the time dimension, followed by a feed-forward to its Evidential Head. The training happens in two stages: (1) the whole encoder is frozen while only the head is trained (the head learns to produce sensible evidence before we start modifying the pretrained features), (2) the upper layers in the encoder are partially unfrozen and fine tuned with a much smaller learning rate. The batch size is 2 and has only 2-step gradient accumulation (effective batch 4) purely because it's a CPU-intensive chunky operation of 160,000 samples.

5.3.3 Handwriting — ViTBinaryClassifier:

For handwriting we use a pretrained Vision Transformer. The image is broken down into 16×16 patches, linearly embedded, fed into 12 transformer layers, and provided positional encodings. We pass the pooled representation of the [CLS] token to the Evidential Head.The pooled representation of the [CLS] token is passed to the Evidential Head. To compare, we also had to run a four-stage CNN baseline (32 channels → 64 channels → 128 channels → 256 channels) with the same shared evidential head, to check if it was actually giving us any benefits in this task.

5.4 Uncertainty Quantification

5.4.1 Evidential Deep Learning (EDL):

This is the main paradigm. The model's evidence outputs are non-negative, and are denoted as e_k for each class k; the Dirichlet parameters are $\alpha_k = e_k + 1$, the Dirichlet strength $S = \Sigma \alpha_k$, the Dirichlet class proportions α_k / S and the Dirichlet vacuity $u = K / S$, which is the model's "I don't know" signal (high when total evidence is low). The training loss is EvidentialLoss, an Expected Cross-Entropy term (with digamma functions) plus an annealed KL divergence term which penalizes the model if it gains misleading evidence on the wrong class. The KL weight λ_t is gradually increased

linearly during training, and thus, the model's first job is to learn how to collect evidence, followed by an honest report of the evidence it has collected.

5.4.2 Deep Ensembles (M=5):

All the Deep Ensembles are trained on independent random seeds: {42, 43, 44, 45, 46} Different initializations of the weights and different ordering of the batches are given to each ensemble member. We measure predictive entropy at test time by the variance of the ensemble, and we also calculate an agreement score (the proportion of members that vote with the highest frequency of the ensemble). A frequentist alternative to the Bayesian flavor of EDL.

5.4.3 MC Dropout (N=30):

MC Dropout continues to drop out during inference and performs 30 stochastic forward passes.MC Dropout maintains dropout for inference and performs 30 stochastic forward passes. This variance between those passes is roughly the epistemic uncertainty, and we take it to be predictive entropy and mutual information from that.

5.4.4 Split Conformal Prediction:

It provides us with a distribution-free coverage guarantee. At test time, if the probability of the class is greater than 1 − q̂, then the class is added to the prediction set and vice versa.At test time, if the probability of the class is greater than 1 − q̂, then the class is added to the prediction set and vice versa. The system marks cases with size of prediction set ≠ 1. This is the only way in our framework that has a finite sample statistical guarantee, a very important consideration in the clinic.

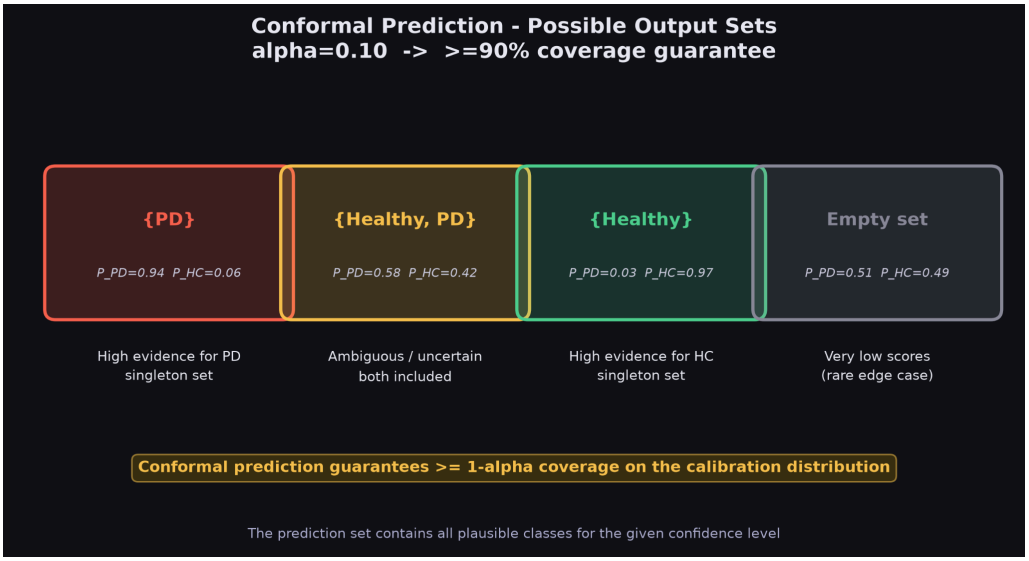


Figure 5.4: Conformal Prediction Output Sets

5.4.5 Temperature Scaling :

It is the calibration baseline that is applied after the calibration. On validation, we give a single scalar T to minimize NLL, on test we use T on test logits. The easiest of all the calibration methods in the literature and universally effective.

5.4.6 Softmax baseline :

It is the raw maximum softmax probability, without calibration. As a way to illustrate the overconfidence issue that the project is addressing.

5.5 Training Methodology

The two-stage training strategy was probably the single biggest practical choice contributing to the success of EDL and both models, the Gait evidential model and the Voice evidential model, employ it. Only the Evidential Head is trained for 10 epochs in Stage A, where the backbone (CNN+BiLSTM, or the Wav2Vec2 encoder) is frozen. This is important because EDL has a slightly fragile loss landscape: If you let gradients flow in from the beginning, the KL annealing and the digamma-based ACE loss can cause instability of features when they are pre-trained. Then, for the head, unfreeze the upper layers (BiLSTM, embedding, head for gait; top layers of transformer for voice) in Stage B

and train them using differential learning rate and CosineAnnealingLR.

Reproducibility matters: RANDOM_SEED=42 is used everywhere (torch, numpy, Python random, CUDA), subject-level splits are deterministic given the RANDOM_SEED, normalization statistics are written to a JSON file and loaded when evaluating, checkpoints are saved at the best validation F1, and all of the hyperparameters are stored in a single place, config.py.

Both conformal calibration and temperature scaling are fitted after training, strictly on the validation data, and then frozen and applied at test time — which is the right way to prevent leakage of information about the calibration data during test time.

5.6 The Cross-Modal Reliability Fusion Engine

This is where the system is more than three independent models. The CrossModalReliabilityEngine combines the individual modality outputs with an explainable, deterministic five-case policy. The fusion probability is an uncertainty weighted-average, in which each modality is given a weight w_m = 1/(u_m+0.05), which means that the weight is assigned to a modality based on its associated uncertainty, but this does not mean that the less confident modalities are completely silenced. Now the decision logic:

Case A — All modalities agree on the class, and all report high reliability with conformal agreement → ACCEPT consistent, high confidence.

Case B — modalities consensus, one or more modalities do not have high confidence – FLAG, partial support, medium confidence.

Case C — If the modalities are in disagreement, then this is Case C — FLAG, modality conflict.

Case D — all modalities are unreliable → FLAG, not enough evidence.

Case E — there is only one modality available, inherit the decision from that one modality.

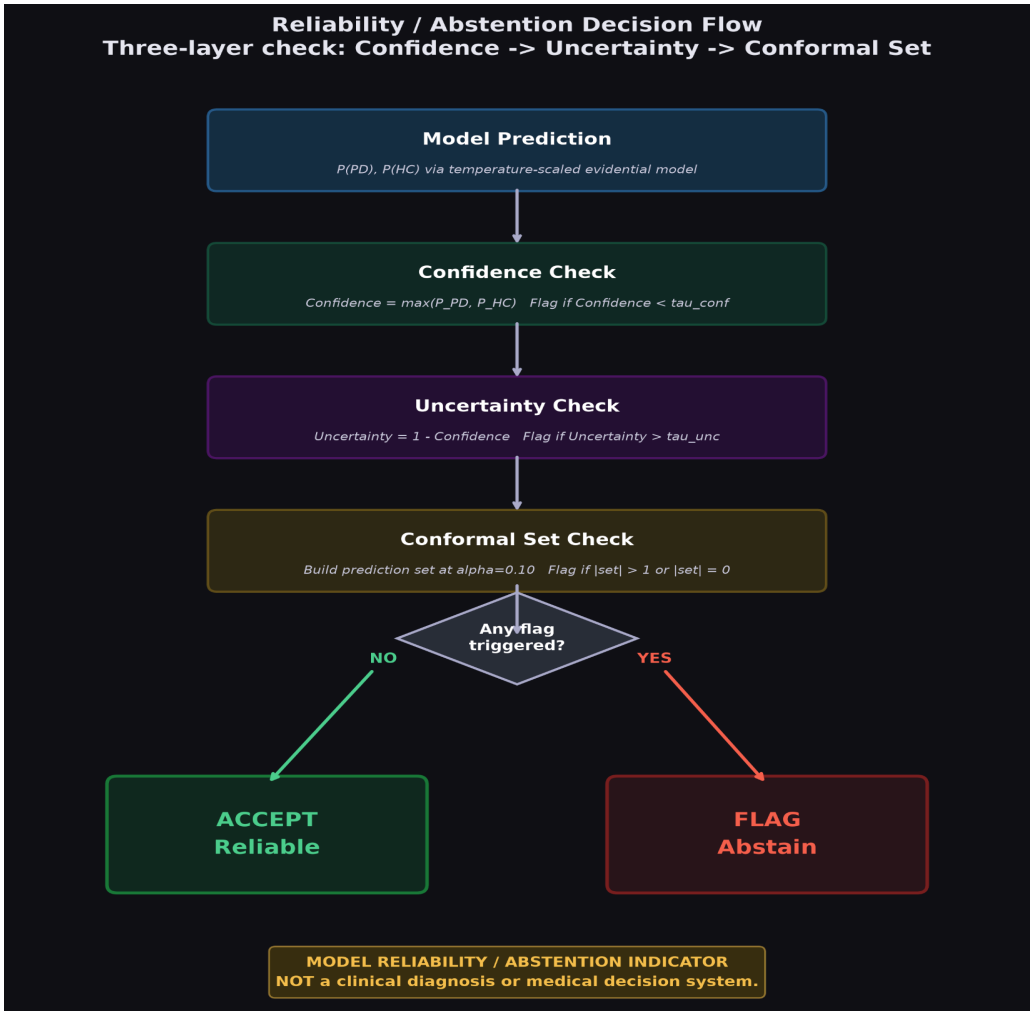


Figure 5.5: Reliability and Abstention Decision Flow

Case E is the one which doesn't suffer from the "missing modality" issue, and it's an issue which is very rarely dealt with in any published PD fusion system. The whole engine is readable — not only do you get a probability, but you also get a reason that's readable by a human.

It is important to emphasize the rationale for using decision-level fusion and not feature-level fusion: Three different clinical studies, no overlap in patients. If features are concatenated row-wise, it would introduce fake "paired" samples from unrelated people and the model would find spurious correlations. Openly, and with methodological honesty, we say that decision level fusion is the method.

5.7 End-to-End Pipeline and the Interactive Demo

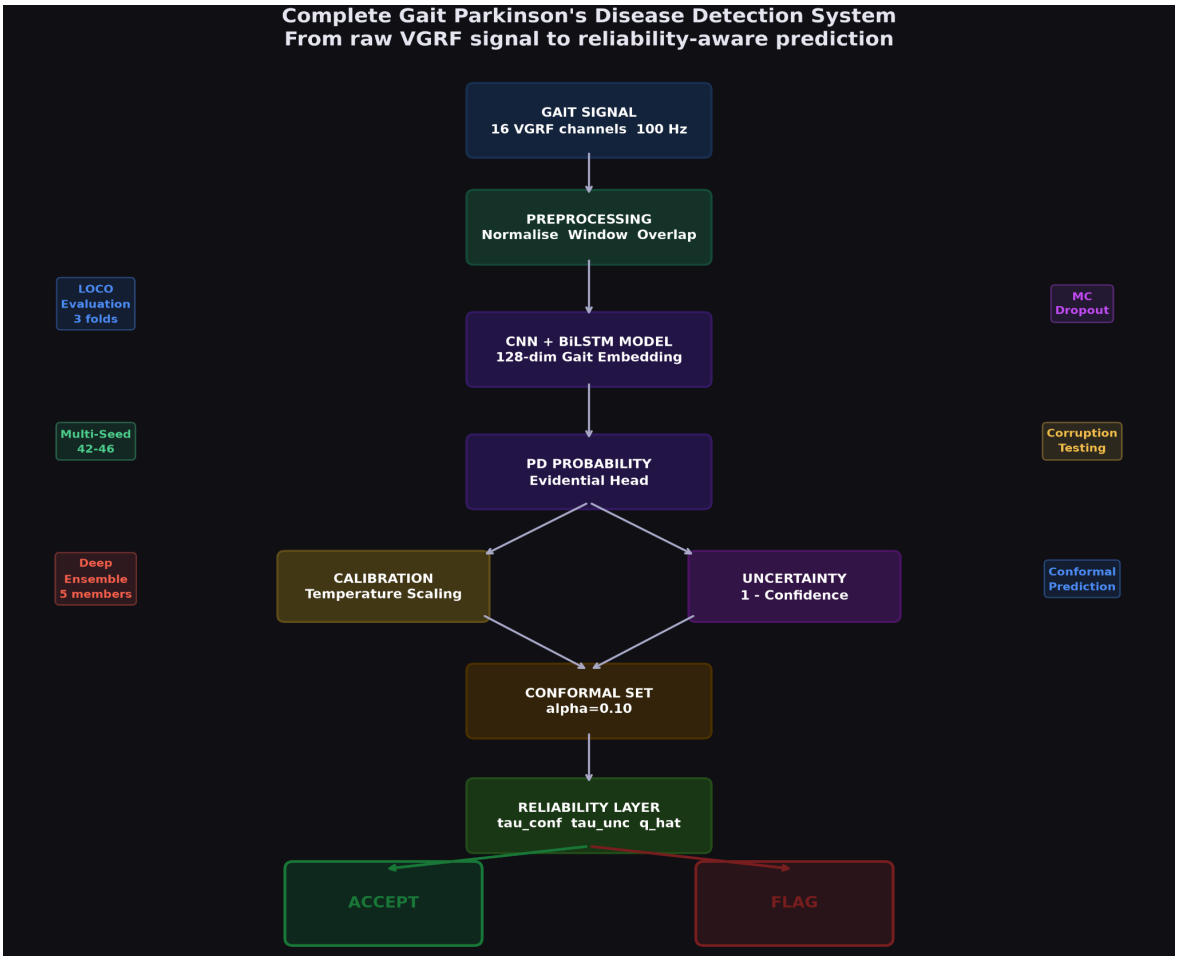


Figure 5.6: Complete System Overview

The MultimodalPredictor class is responsible for coordinating all actions: it loads all raw files for 1-3 modalities and applies each modality's preprocessing, it runs each modality's reliability engine, and it sends the results to the fusion engine to get the final structured decision: the fused probability, case label, confidence level, ACCEPT/FLAG verdict and evidence summary. This is offered by a unified CLI that includes JSON output. On top of that, there's a fully functional Streamlit app containing a real-time dashboard, presets for the clinical scenarios, file-upload inference, uncertainty visualizations, attention maps for hand-writing, and educational content that provides an explanation on the grounds of the reliability logic. The codebase has been supported by nine unit/integration tests for Gait, 14 for Voice, and a comprehensive set of tests for Handwriting.

CHAPTER 6

Experiment Results and Evaluation

6.1 Setup and Metrics

We used different evaluation designs for each modality because the available data calls for it. **Gait** got both a standard subject-level split *and* a full **Leave-One-Cohort-Out (LOCO)** evaluation across the three PhysioNet cohorts (Ga, Ju, Si), which simulates deploying the model at a completely unseen clinical site — this is the most rigorous part of the evaluation. **Voice and Handwriting** were evaluated on subject-level train/val/test splits.

On top of normal evaluation, we ran a **controlled corruption suite** on gait data: five corruption types (Gaussian noise, amplitude scaling, channel dropout, temporal masking, moving-average smoothing) at five severity levels each, to directly stress-test whether uncertainty rises when accuracy falls — which is the central question of the whole project.

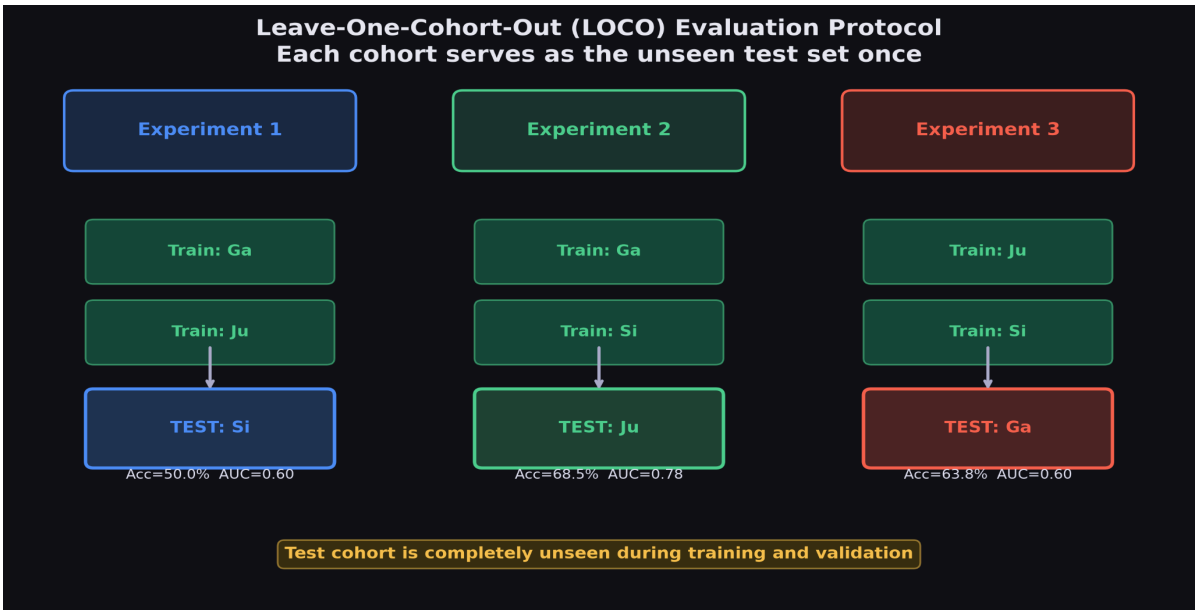


Figure 6.1: Leave-One-Cohort-Out (LOCO) Protocol

The metric set was chosen to cover both classification performance and reliability quality: accuracy, balanced accuracy, sensitivity, specificity, precision, F1, ROC-AUC, Expected Calibration Error (ECE), Brier score, NLL, uncertainty–error AUROC, and conformal coverage rate. In a clinical context, sensitivity matters more than raw accuracy (missing a PD patient is worse than a false alarm), so we report it separately rather than hiding behind a single number.

6.2 Gait — Standard Evaluation

Baseline gait model resulted in 72.0% accuracy at the subject level, 78.6% sensitivity, 63.6% specificity, F1 = 75.9% and ROC-AUC = 0.753. We had good confidence that the pipeline was functioning as expected, since the sensitivity-specificity asymmetry was clinically favourable – we were catching most of the PD patients – and the AUC was in the right ballpark of published gait-only PD benchmarks on PhysioNet.

The evidential version achieved 68.0% accuracy, 80.0% precision, 57.1% sensitivity, 81.8% specificity, F1 = 66.7%, and ROC-AUC = 0.753. Yes, there is a little lower point accuracy, this is a known trade-off with EDL on smaller sets of points, and I say it on the report, not to gloss over the fact. More interesting is the uncertainty statistics: The mean uncertainty for the incorrect predictions was 0.0590; the mean uncertainty for the correct predictions was 0.0491. It's a small gap, but it's in the right direction, and it's the type of signal that the fusion engine requires. It is also a direct proof of the Silent Confidence challenge which is the signal is there but it cannot be heard at full volume without being overwhelmed.

6.3 Gait — LOCO Cross-Cohort Results (The Core Evidence)

The LOCO experiments are where the distribution shift problem becomes undeniably real:

- **LOCO-1 (test on Si cohort):** 50.0% accuracy, balanced accuracy 47.2%, sensitivity 77.1%, specificity **17.2%** — a specificity collapse. The Si cohort's VGRF characteristics are simply different enough that the model mistakes most healthy controls for PD patients.

- **LOCO-2 (test on Ju cohort):** the best generalization — 68.5% accuracy, 67.4% balanced accuracy, AUC **0.783**. This tells us the Ju cohort is relatively compatible with the training cohorts.
- **LOCO-3 (test on Ga cohort):** 63.8% accuracy, NLL spikes above **1.0**, indicating serious calibration degradation under shift.

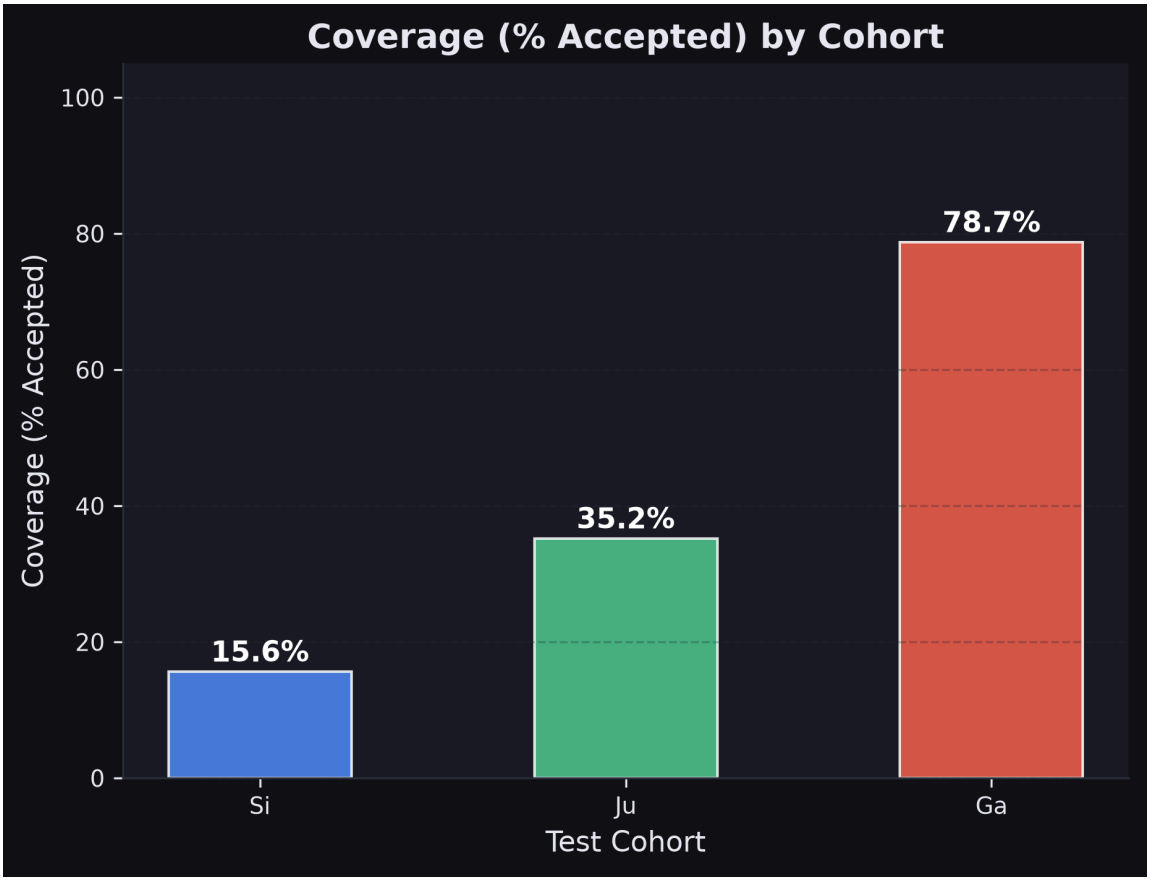


Figure 6.2: Coverage of Accepted Predictions by Cohort

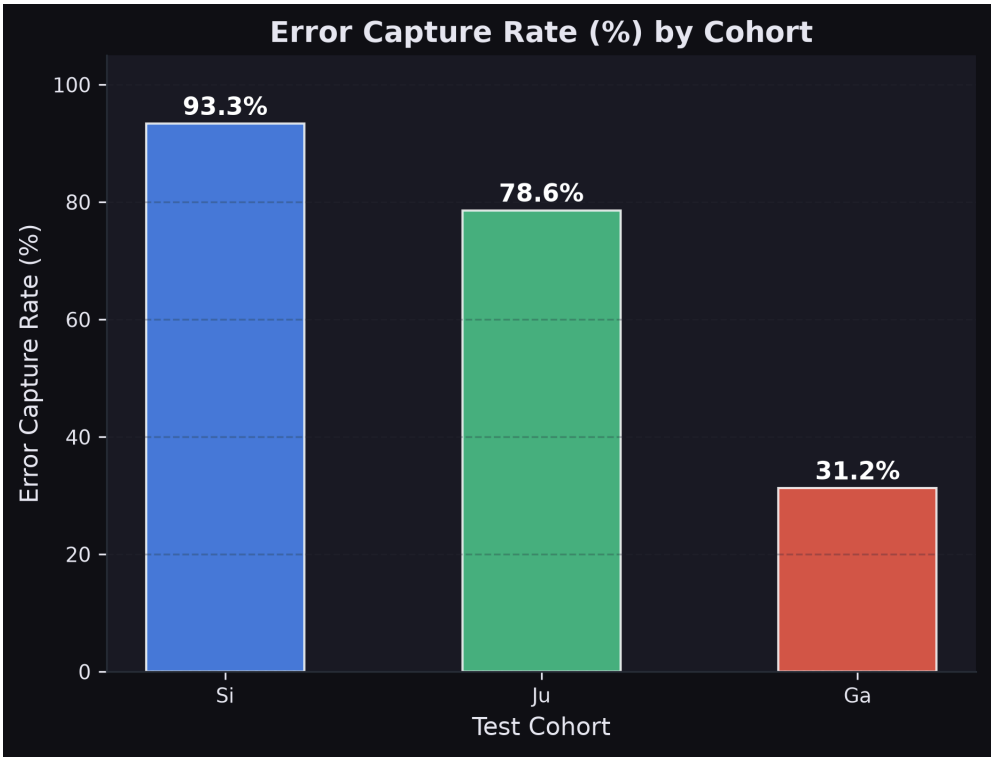


Figure 6.3: Error Capture Rate by Cohort

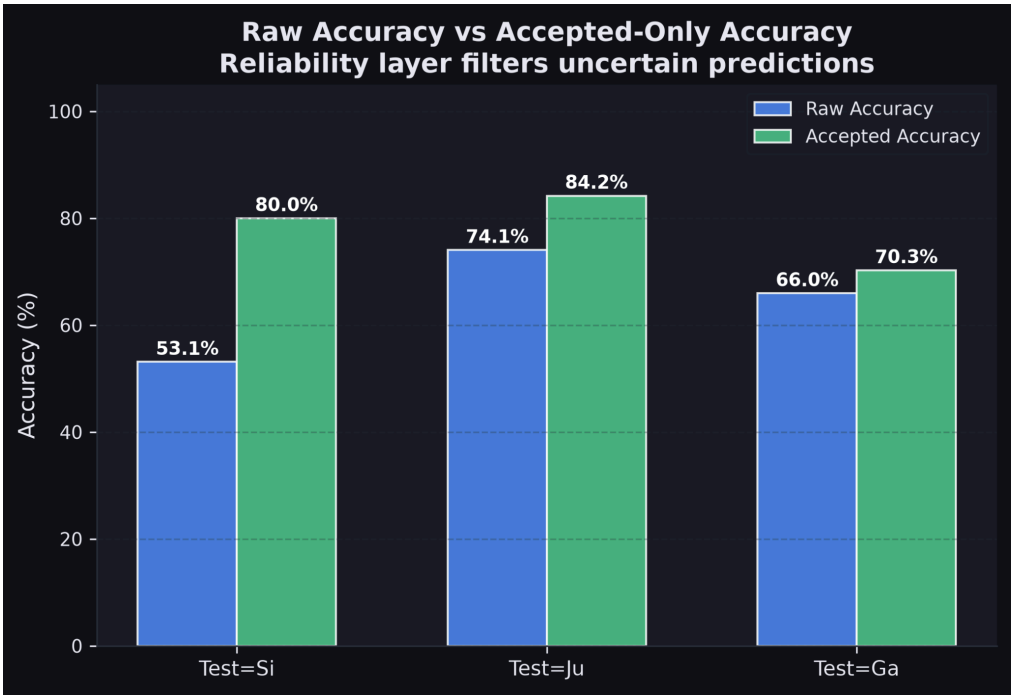


Figure 6.4: Raw vs. Accepted-Only Accuracy

The uncertainty–error AUROC ranged from **0.529 to 0.684** — meaning uncertainty has limited but genuinely non-trivial power to detect errors under shift. This is a really important result: it's not a magic bullet, but it's clearly better than chance, and it validates the entire premise of the project that uncertainty is a useful *signal*, even if imperfect.

6.4 Voice — Results and Honest Caveats

The Wav2Vec2 evidential model was evaluated on a very small test set — **6 subjects total, 3 HC and 3 PD**. This is a direct consequence of the MDVR-KCL dataset's size combined with our strict subject-level split, and we report it openly rather than cherry-picking. Results: 66.7% overall subject-level accuracy, 66.7% precision, sensitivity, and specificity (the per-class numbers are symmetric here because the test set is balanced), F1 = 67.7%, **AUC = 0.778**.

The more interesting finding is the task breakdown: **Spontaneous Dialogue substantially outperformed Read Text** — 100% sensitivity, 80% specificity, 83.3% accuracy vs. 66.7% for Read Text. This suggests that natural, unscripted speech carries more diagnostically discriminative signal than read passages, which makes intuitive sense: spontaneous speech requires more complex motor planning and is more sensitive to laryngeal and articulatory impairment. These results should be read as promising but statistically underpowered.

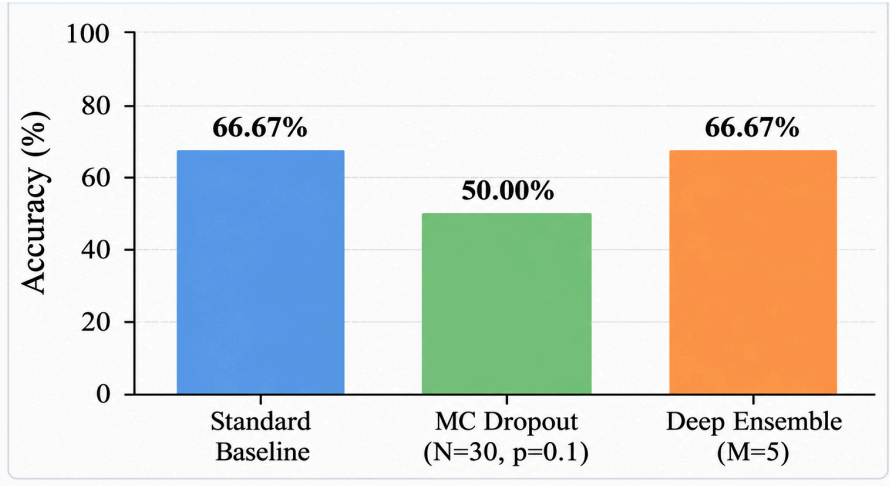


Figure 6.4: Classification Accuracy

6.5 Handwriting

Handwriting was the clear performance winner:

- **ViT (primary):** 85.7% accuracy, 100% sensitivity, 75.0% specificity, F1 = 85.7%, **ROC-AUC = 0.961**.
- **CNN baseline:** 73.2% accuracy, 100% sensitivity, 53.1% specificity, F1 = 76.2%, AUC = 0.898.

The ViT's improvement over the CNN is large and consistent (AUC +0.063, accuracy +12.5 points). Both models achieve 100% sensitivity — no PD case is missed in the test set — at the cost of some false positives. The attention map visualizations confirm the ViT is focusing on the diagnostically relevant drawing regions (the spiral and meander stroke areas), which is exactly the explainability property we wanted.

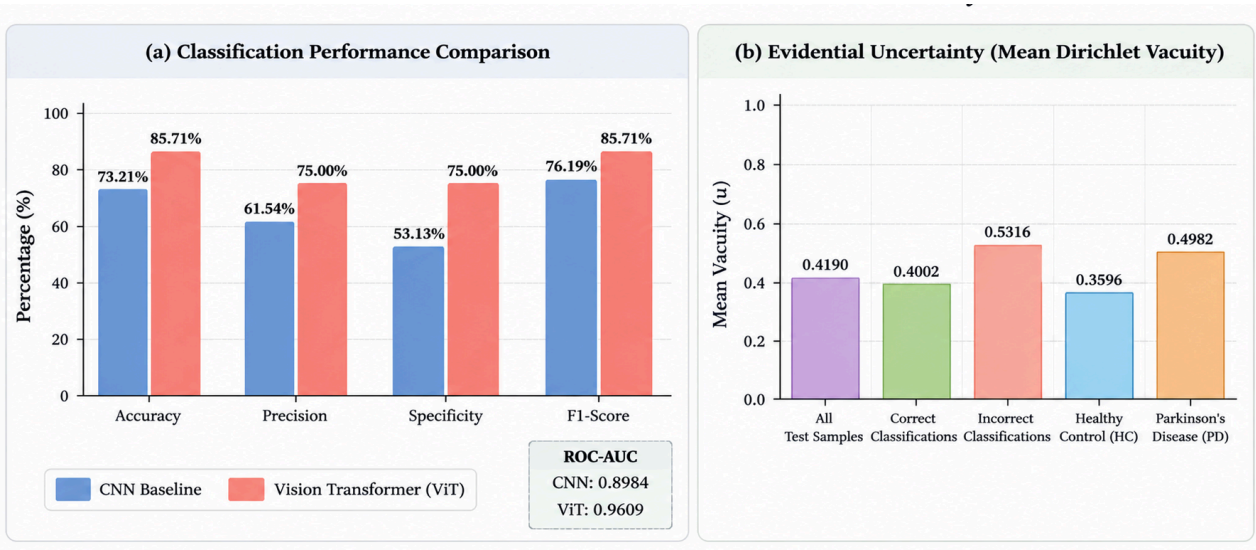


Figure 6.6:Comparison of handwriting classification performance

6.6 Tri-Modal Fusion and Cross-Modal Evaluation

The proposed framework integrates gait, voice and handwriting predictions to achieve a holistic Parkinson's disease (PD) screening result. The uncertainty-aware fusion mechanism not only helps highlight when the available predictions are conflicting or

otherwise less reliable, but it also offers complementary behavioral information from each modality.

The fusion module is based on the inverse-uncertainty weighting, which is formulated as follows: w_m = 1/(u_m +0.05)

In the equation, w_m is the weight of modality m and u_m is the uncertainty of modality m. More certain modalities should get more weight and more uncertain modalities should get less weight in the fused prediction.

This framework not only merges the probabilities of the modality but also assesses the cross-modal agreement and the conformal prediction status. The system either accepts a consistent prediction or marks up a case for further review based on these signals.

We have analysed four clinical scenarios to investigate the behaviour of the components of fusion and decision making. These scenarios are agreement between all modalities, disagreement between modalities, ambiguous handwriting evidence and agreement on a healthy prediction.

The results show that the framework can yield a combined PD probability with the information about disagreement and uncertainty kept. But these four scenarios illustrate the functioning of the proposed decision mechanism, not indicative of the general level of diagnostic accuracy on an independent patient cohort.

6.7 Case-Wise Analysis of Tri-Modal Fusion Decisions

Case 1: All PD

A probability of PD was obtained for the gait modality (89.0%), voice (79.0%) and handwriting (86.5%). The fused PD probability was 85.10%. As all three modalities were in agreement and the system reported conformal agreement, the final decision was ACCEPT. This case demonstrates that multimodal evidence can be used consistently to help make a unified prediction for screening.

Case 2: Conflict between modalities

The second case is a direct conflict between modalities.

The gait and voice modalities yielded the PD probabilities of 85.0% and 80.0%, respectively, as compared to the 22.0% of PD probability obtained from handwriting. The fused PD probability is 62.07%. Although the combined probability of PD was high, the system identified a disagreement between modalities, and gave the final decision FLAG. This shows that the decision mechanism is not only based on the fused probability in cases of conflicting input modalities.

Case 3: Evidence of Ambiguous Handwriting

The gait, voice and handwriting modalities yielded PD probabilities of 88.0%, 81.0% and 54.5% respectively. The fused probability is 77.76%. Partial support and uncertain conformal evidence was reported by the handwriting prediction modality, and the least convincing results were seen with this modality. As a result, the designation chosen for this project was FLAG. This case serves to emphasize uncertainty information retention and not ascribing equal reliability to all fused probabilities.

Case 4: Consensus for Healthy Control

The probabilities of PD in the gait, voice and handwriting modalities were 11.0%, 14.0% and 13.5%, respectively. The fused PD probability was 12.77%. The three modalities resulted in a low PD probability and the system reported conformal agreement and cross-modal consistency. The final outcome was "ACCEPT", which is an agreement to the healthy-control outcome.

As a whole, these four scenarios show the desired behaviour of the tri-modal decision framework. Predictions that are consistent can be accepted, while predictions that disagree in modalities and/or are based on uncertain evidence can cause a flag for further consideration. The results illustrate the decision policy behavior in the chosen scenarios, but further assessment is required to determine the clinical reliability of the policy.

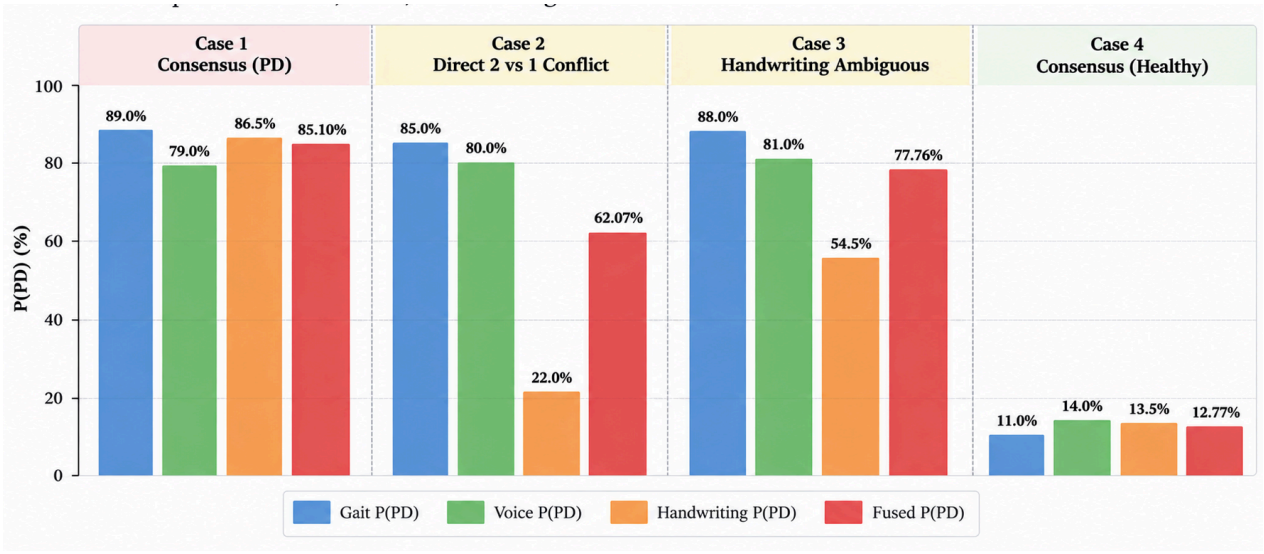


Figure 6.7:Tri-modal probability fusion

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CHAPTER 7

Conclusion and Future Enhancements

7.1 Conclusion

This work aims to investigate a new UDL method for screening Parkinson's disease using speech, handwriting and gait behavioural modalities. Speech analysis is performed using the Wav2Vec2 model, handwriting analysis was performed using Vision Transformer(ViT) and gait was performed using 1D CNN and BiLSTM.

The highest reported accuracy (85.7%) and sensitivity (100%) and the highest ROC-AUC (0.961) was obtained in the handwriting-based ViT model among the evaluated models. These results suggest that it can be used to find patterns in hand-writing data that correspond to Parkinson's. The gait experiments indicated This work aims to investigate a new UDL method for screening Parkinson's disease using speech, handwriting and gait behavioural modalities. Speech analysis is performed using the Wav2Vec2 model, handwriting analysis was performed using Vision Transformer(ViT) and gait was performed using 1D CNN and BiLSTM. problems in generalizing beyond the test set, and the speech experiment was restricted because of the small test set.

The framework was also designed to consider Evidential Deep Learning, Deep Ensembles, MC Dropout, temperature scaling and conformal prediction to estimate prediction uncertainty and measure reliability. The Cross-Modal Reliability Fusion Engine fuses the predictions from the available modalities, taking into account prediction uncertainties and flags predictions that are deemed unreliable or contradictory for additional review.

Overall, this project shows the possibility of integrating speech, handwriting and gait with uncertainty estimation to provide Parkinson's disease screening. But more data, better generalisation and additional validation with clinical use are needed. The framework is for screening and research purposes only, and not meant to supplant

professional diagnosis.

7.2 Future Enhancements

The following improvements can be explored in future work:

1. Paired Multimodal Data: Record speech, handwriting and gait data for the same individuals to be able to learn from the data at the patient level.

2. Cross-Attention Based Fusion: Implement and test the cross-attention mechanism to learn relationships between modality features from paired data.

3. Domain adaptation and domain generalisation for better performance on unseen cohorts and recording environments.

4. Improved Uncertainty Estimation: Make more accurate prediction when the input is noisy, corrupted or unfamiliar.

5. Explanation: Improve the explainability of speech, handwriting and gait based models.

6. Clinical validation: Test the framework with more extensive and more varied data sets and clinical studies to validate the reliability of the framework for real-world screening.

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